FUN-6.D.3 Techniques for finding antiderivatives include rearrangements into equivalent forms, such as long division and completing the square.

6-10 Integrating Functions Using Long Division and Completing the Square

Get Ready...

Evaluate each integral:

A.) $\int \frac{8x}{\sqrt{1-5x^2}}\,dx =$

$u = 1-5x^2$

$du = -10x\,dx$

$\frac{8}{-10}\int \frac{-10x\,dx}{\sqrt{1-5x^2}} = \frac{-4}{5}\int \frac{1}{\sqrt{u}}\,du = \frac{-4}{5}\int u^{-1/2}\,du$

$= -\frac{4}{5}\cdot 2u^{1/2} + C = \boxed{-\frac{8}{5}(1-5x^2)^{1/2} + C}$

B.) $\int \frac{8}{\sqrt{\boxed{1}-\boxed{5x^2}}}\,dx =$

$\int \frac{du}{\sqrt{a^2-u^2}} = \arcsin\left(\frac{u}{a}\right) + C$

$u^2 = 5x^2$

$u = x\sqrt{5}$

$du = \sqrt{5}\,dx$

$a^2 = 1$

$a = 1$

$\frac{8}{\sqrt{5}}\int \frac{1\sqrt{5}}{\sqrt{1-5x^2}}\,dx = \boxed{\frac{8}{\sqrt{5}}\arcsin(x\sqrt{5}) + C}$

Sometimes we must simplify the integrand before evaluating an integral. This may involve several types of algebraic manipulation. Let's look at the algebraic techniques of long division and completing the square.

Long division:

Example 1: $\int \frac{x^3 - 4x^2 - 4x + 20}{x^2 - 5}\, dx =$

$$\int \left(x - 4 + \frac{x}{x^2-5}\right) dx$$

$$= \boxed{\frac{1}{2}x^2 - 4x + \frac{1}{2}\ln\left|x^2-5\right| + C}$$

$$\begin{array}{r} x - 4 + \frac{x}{x^2-5} \\ x^2-5 \overline{\big) x^3 - 4x^2 - 4x + 20} \\ -x^3 \qquad + 5x \\ \hline -4x^2 + x + 20 \\ +4x^2 \qquad \mp 20 \\ \hline x \end{array}$$

$$\frac{1}{2}\int \frac{2x}{x^2-5}\, dx \qquad u = x^2 - 5$$

$$du = 2x\,dx$$

$$= \frac{1}{2}\int \frac{1}{u}\, du = \frac{1}{2}\ln|u| + C$$

Example 2: $\displaystyle\int \frac{3x^3 - x^2 - 5x + 1}{x - 2}\,dx =$

$x-2=0$

$$\begin{array}{r|rrrr} 2 & 3 & -1 & -5 & 1 \\ & \downarrow & 6 & 10 & 10 \\ \hline & 3 & 5 & 5 & \boxed{11} \\ & x^2 & x & \text{const} & \text{Remainder} \end{array}$$

$$= \int \left(3x^2 + 5x + 5 + \frac{11}{x-2}\right) dx$$

$$= x^3 + \frac{5}{2}x^2 + 5x + 11\ln|x-2| + C$$

$$\int \frac{11}{x-2}\,dx \qquad u = x-2, \quad du = dx$$

$$= 11\int \frac{1}{x-2}\,dx$$

$$= 11\int \frac{1}{u}\,du = 11\ln|u| + C$$

Completing the Square:

Example 3: $\int \frac{3}{x^2 - 4x + 9}\, dx =$

$x^2 - 4x + 9$

$\underbrace{x^2 - 4x + 4} + 9 - 4$

$(x-2)^2 + 5$

$= \int \frac{3}{\boxed{(x-2)^2} + \boxed{5}}\, dx$

$u^2 = (x-2)^2$ $\quad a^2 = 5$

$u = x - 2$ $\quad a = \sqrt{5}$

$du = dx$

$\int \frac{du}{a^2 + u^2} = \frac{1}{a} \arctan\left(\frac{u}{a}\right) + C$

$= 3 \int \frac{1\, dx}{(x-2)^2 + 5} = 3 \cdot \frac{1}{\sqrt{5}} \arctan\left(\frac{x-2}{\sqrt{5}}\right) + C$

$= \frac{3}{\sqrt{5}} \arctan\left(\frac{x-2}{\sqrt{5}}\right) + C$

Example 4: $\int \frac{3}{\sqrt{-4x^2-16x-7}}\,dx =$

$-4x^2-16x-7$

$-4(x^2+4x+4)-7+16$

$-4(x+2)^2+9$

$$\int \frac{3}{\sqrt{\boxed{9}-\boxed{4(x+2)^2}}}\,dx$$

$u^2=4(x+2)^2$

$u=2(x+2)$

$2x+4$

$du=2dx$

$a^2=9$

$a=3$

$$\int \frac{du}{\sqrt{a^2-u^2}} = \arcsin\left(\frac{u}{a}\right)+C$$

$$\frac{3}{2}\int \frac{2\,dx}{\sqrt{9-4(x+2)^2}} = \boxed{\frac{3}{2}\arcsin\left(\frac{2(x+2)}{3}\right)+C}$$